MIL-L-51292C(EA)
25 September 1987
SUPERSEDING
MIL-L-51292B(MU)
9 June 1970

MILITARY SPECIFICATION

LAUNCHER AND GRENADES, SMOKE: HC AND WP, M176

This specification is approved for use within U.S. Army Chemical Research, Development and Engineering Center, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers one type of launcher for dispensing grenades.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

FEDERAL

NN-P-530	-	Plywood, Flat Panel.
TT-W-572	-	Wood Preservative: Water-Repellant.
PPP-B-621	-	Boxes, Wood, Nailed and Lock-Corner.
PPP-F-320	-	Fiberboard: Corrugated and Solid, Sheet Stock (Container Grade) and Cut Shapes.
PPP-T-66	-	Tape, Packaging, Vinyl Plastic Film.

: Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed: to: Commander, U.S. Army Chemical Research, Development and Engineering Center, ATTN: SMCCR-SPT-S, Aberdeen Proving Ground, MD 21010-5423 by using: the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter. :

AMSC N/A A FSC 1330

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MILITARY

MIL-C-2439 - Containers, Ammunition, Fiber, Spirally Wound.
MIL-G-12327 - Grenade, Hand, Smoke, HC, AN-M8.

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.
MIL-STD-709 - Ammunition Color Coding.
MIL-STD-1168 - Ammunition Lot Numbering.

2.1.2 Drawings and publications. The following drawings and publications form a part of this specification to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of the solicitation.

DRAWINGS

U.S. ARMY CHEMICAL RESEARCH, DEVELOPMENT AND ENGINEERING CENTER

13-12-1 - Launcher and Grenades, Smoke: HC and WP, M176.
13-12-19 - Packing for Grenade Launcher, M176.
13-12-20 - Marking Diagram for Grenade Launcher, M176.
13-12-21 - Palletization for Grenade Launcher, M176.

PUBLICATIONS

U.S. ARMY CHEMICAL RESEARCH, DEVELOPMENT AND ENGINEERING CENTER

R14-9 - Inspection Standard for Fluoroscopic or Radiographic Evaluation of; Launcher and Grenade, HC and WP, XM176.

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specifications, specification sheets or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

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3. REQUIREMENTS

3.1 Materials and components.

3.1.1 Materials. All materials cited on Drawing 13-12-1 and on the subsidiary drawings shall conform to the specifications thereon or to the specific characteristics set forth on the drawings.

3.1.2 Components. All components shall conform to the specifications and drawings listed on Drawing 13-12-1 and subsidiary drawings, as applicable. The AN-M8 grenades produced more than 90 days prior to assembly into launchers shall be tested within 90 days prior to assembly into launchers and meet the functional requirements of MIL-G-12327.

3.2 Assembly. The launcher shall be assembled as shown on Drawing 13-12-1.

3.3 Radiographic or fluoroscopic evaluation. The launcher shall be radiographed or fluoroscoped for missing components, in accordance with 4.4.4.1.

3.4 Leakage. The launcher shall not leak air when submerged in water at a temperature of 170° ± 10°F, for a period of 30 to 35 seconds when tested in accordance with 4.4.4.2.

3.5 Functioning. The launcher shall eject the grenades to an average impact distance of 100 to 170 feet based on the average of the entire sample that functioned. However, no grenade shall impact at a distance of less than 80 feet. In addition, the two components of the sabot (holder and cover) shall separate on or before impact and the grenades shall ignite and emit a smoke cloud. Angle of elevation for firing test stand shall be 33 ± 1/2 degrees above the horizontal, and the front underside of the launcher shall be 77 ± 1 inch above the average impact area, 100 to 170 feet down range. Test shall be in accordance with 4.4.4.3.

3.6 Low temperature. The launcher shall function subsequent to a conditioning period of 48 ± 1 hours at a temperature of -25° ± 2°F, when tested in accordance with 4.3.3.1.

3.7 High temperature. The launcher shall function subsequent to a conditioning period of 48 ± 1 hours at a temperature of 125° ± 5°F, when tested in accordance with 4.3.3.2.

3.8 Water immersion. The launcher shall function subsequent to being submerged in water at ambient temperature for 60 ± 5 minutes and then removed from the water and fired after 24 to 25 hours, in accordance with 4.3.3.3.

3.9 Preproduction. Prior to the start of regular production, a preproduction sample of launchers shall be produced in accordance with this specification for examination and test (see 4.3).

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3.10 Workmanship. The launcher shall be free from foreign matter and damage which could affect functioning.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.1.2 Objective evidence. The contractor shall provide objective evidence acceptable to the contracting officer that the requirements of 3.1 and section 5 for which specific inspection has not been provided in this specification have been satisfied.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

(a) Preproduction inspection (see 4.3).
(b) Quality conformance inspection (see 4.4).

4.3 Preproduction inspection

4.3.1 Sample. A preproduction sample of 30 launchers shall be manufactured using the same methods, materials, equipment and processes as will be used during regular production.

4.3.2 Inspection procedures.

4.3.2.1 For examination. The preproduction sample launchers shall be examined in accordance with the classification of defects, the AQLs are not applicable.

4.3.2.2 For destructive test. Sample launchers shall be tested in accordance with table I.

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TABLE I. Preproduction

Quantity	Requirements	Test	Defect to be observed
10	3.6 low temperature	4.3.3.1 and 4.4.4.3	Duds, misfires, short range, or sabot fails to separate
10	3.7 high temperature	4.3.3.2 and 4.4.4.3	
10	3.8 water immersion	4.3.3.3 and 4.4.4.3	

4.3.3 Tests.

4.3.3.1 Low temperature. The launcher shall be placed in a cold chamber at the specified temperature for the required period of time, and fired 5 to 6 minutes after removal from the chamber, or insulated transport container.

4.3.3.2 High temperature. The launcher shall be placed in a hot chamber at the specified temperature for the required period of time and fired 5 to 6 minutes after removal from the chamber, or insulated transport container.

4.3.3.3 Water immersion. The launcher shall be submerged in water to a depth of 6 to 7 inches, at the specified temperature for the required period of time and then stored for the required period of time.

4.3.4 Rejection. If any assembly, component, or test specimen fails to comply with any of the applicable requirements, the preproduction sample shall be rejected, with the exception of dud grenades. The preproduction lot shall be acceptable if it contains a total of three or less dud grenades. The Government reserves the right to terminate inspection upon any failure of any assembly, component or test specimen to comply with any of the requirements. The contractor shall obtain written approval from the contracting officer prior to proceeding with regular production.

4.4 Quality conformance inspection.

4.4.1 Lotting. Each lot shall be identified and controlled in accordance with MIL-STD-1168. Contained in any one interfix number of launchers, no more than one interfix number from one manufacture of the following is permitted:

(a) M34 grenades
(b) M8 grenades
(c) M225-propellant cartridges

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4.4.2 Sampling.

4.4.2.1 For examination and radiographic or fluoroscopic evaluation. Sampling shall be conducted in accordance with MIL-STD-105.

4.4.2.2 For destructive test. Sampling shall be conducted in accordance with MIL-STD-105, level S-3, and an AQL of 4.0, and shall be taken from the sample used for examination (see 4.4.2.1).

4.4.3 Inspection procedure.

4.4.3.1 For examination. Sample launchers and the packaging thereof shall be examined in accordance with the classification of defects and MIL-STD-105.

4.4.3.2 For leakage. Each launcher in the lot shall be tested in accordance with 4.4.4.2.

4.4.3.3 For destructive test. Sample launchers shall be tested in accordance with table II and MIL-STD-105, with an AQL of 4.0 percent defective with the following exceptions:

(a) Grenade dud. Launcher sample size of 13: accept 3, reject 4. Launcher sample size of 20: accept 4, reject 5.

(b) Any launcher lot sample which fails to eject the grenades to an average impact distance of 100 to 170 feet shall be cause for rejection.

(c) Any launcher lot sample which has a grenade that impacts at a distance of less than 80 feet shall be cause for rejection.

TABLE II. Functioning

Attribute	Test
Misfires	4.4.4.3
Sabot fails to separate	4.4.4.3

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4.4.3.4 Classification of defects.

(a) Launcher and grenades, smoke: HC and WP, M176 (Drawing 13-12-1.

Categories		Defects	Acceptance standard
Critical:		None defined	
Major:	AQL	0.10 percent defective	
101		Component missing or incorrectly assembled	4.4.4.1
	AQL	1.0 percent defective	
102		Gasket seal missing	
*103		Excessive scoring of cap crimp	
104		Marking incorrect or illegible	
105		Dents in surface of cap	
106		Workmanship (see 3.10)	

*Definite cut or groove on the crimped metal surface of the cap caused by excessive pressure or misalignment of the roller.

(b) Packaging (see section 5).

Categories		Defects
Critical:		None defined
Major:	AQL	4.0 percent defective
101		Loose pack
102		Loose strap
103		Marking incorrect or illegible

4.4.4 Tests.

4.4.4.1 Radiographic or fluoroscopic evaluation. The launcher shall be evaluated in accordance with Inspection Standard R14-9.

4.4.4.2 Leakage. The launcher, in a horizontal position, shall be submerged in water at the specified temperature, to a depth of 1 ± 1/4 inch for the required period of time. The water should contain a wetting agent to prevent bubbles from clinging to the surface of the launcher.

4.4.4.3 Functioning. The launcher shall be fired from the APE 1951-0000 test fixture at the specified elevation. The range shall be measured from the aft end of the launcher.

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5. PACKAGING

5.1 Packaging, level A. Eight grenade launchers shall be packed in accordance with Drawing 13-12-19. The box or the wood parts thereof shall be preserved by completely immersing in a water-repellent wood preservative conforming to type I or II, composition A, of TT-W-572, for a minimum of 3 minutes.

5.2 Marking. In addition to any special marking required by the contract or order, shipping containers shall be marked in accordance with Drawing 13-12-20.

5.3 Palletization. When specified in the contract or order, grenade launchers packed in accordance with 5.1 shall be palletized and marked in accordance with Drawing 13-12-21.

6. NOTES

6.1 Ordering data. Acquisition documents should specify the following:

(a) Title, number, and date of this specification.
(b) The shipment group size and marking, when required.
(c) Palletization when required
(d) Preproduciton.

(1) Time allowed for contractor's submission of samples for Government test and evaluation after award of contract.
(2) Destination of samples when testing is performed by the Government.
(3) Packing, shipping and marking instructions, when applicable.
(4) Time required for the Government to notify the contractor whether to proceed with production.

6.2 Definitions.

(a) Dent: surface irregularity not introduced during normal manufacturing process.
(b) Dud: grenade malfunction due to a mechanical or pyrotechnic failure.
(c) Misfire: launcher does not eject grenades when functioned.

6.3 Subject term (key word) listing.

Grenade
Launcher
White phosphorous

6.4 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-L-51292C(EA)	LAUNCHER AND GRENADES, SMOKE: HC AND WP, M176

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*: _____
b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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